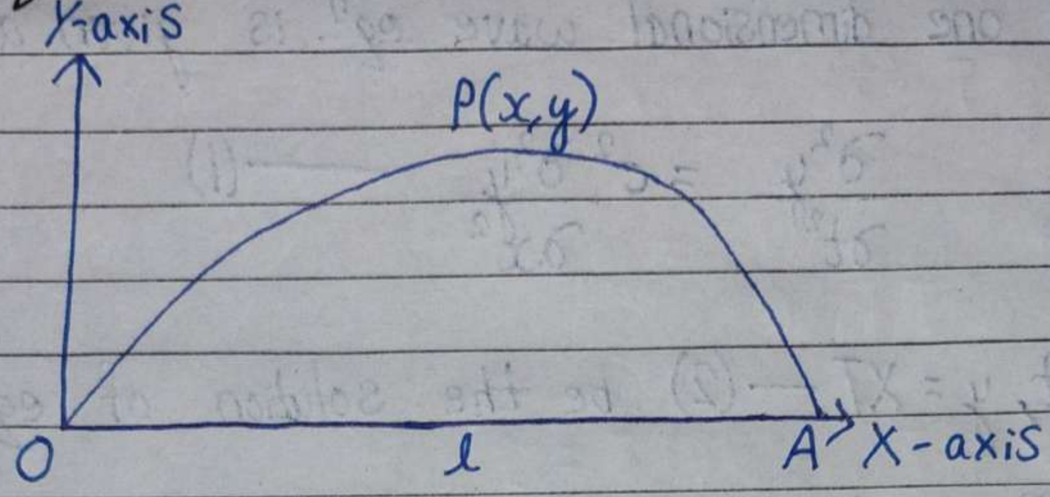
Equation of Vibrating String (One Dimension) Wave Equation



Consider a uniform elastic string of length 'l' streched tightly beween two points O & A and displaced slightly from its equilibrium position OA. Taking the end O as the X-axis and a perpendicular line through O as the Y-axis.

When the string is in motion in the XY-plane, the displacement y of any point of the string is a function of x & t.

Now, the equation of motion for the string is:-

$$\frac{\partial^2 y}{\partial t^2} = c^2 \frac{\partial^2 y}{\partial x^2} \quad \text{[where, c is constant]}$$

which is known as the one dimensional wave equation because the motion is only in vertical direction.

Imp. Solution of Wave Equation

The one dimensional wave eqn. is given as,

$$\frac{\partial^2 y}{\partial t^2} = c^2 \frac{\partial^2 y}{\partial x^2} \quad \text{——} (1)$$

Let, $y = XT$ —(2) be the solution of eqn. (1).

Differentiating eqn. (2) w.r.to x,

$$\frac{\partial^2 y}{\partial x^2} = T \frac{\partial^2 X}{\partial x^2}$$

Again, differentiating eqn. (2) w.r.to t,

$$\frac{\partial^2 y}{\partial t^2} = X \frac{\partial^2 T}{\partial t^2}$$

Putting these two eqn. in eqn. (1),

$$X \frac{\partial^2 T}{\partial t^2} = c^2 T \frac{\partial^2 X}{\partial x^2}$$

Dividing above eqn. by XT,

$$\frac{1}{T} \frac{\partial^2 T}{\partial t^2} = \frac{c^2}{X} \frac{\partial^2 X}{\partial x^2} = K \text{ (say)}$$

Case - I: When $k=0$.

$$\frac{1}{c^2T}\frac{\partial^2 T}{\partial t^2}=0$$

$$\Rightarrow \frac{\partial^2 T}{\partial t^2}=0$$

$$\Rightarrow \iint \frac{\partial^2 T}{\partial t^2}=0$$

$$\Rightarrow \int \frac{\partial T}{\partial t}=C_1$$

$$\Rightarrow T=C_1t+C_2$$

$$\frac{1}{X}\frac{\partial^2 X}{\partial x^2}=0$$

$$\Rightarrow \frac{\partial^2 X}{\partial x^2}=0$$

$$\Rightarrow \iint \frac{\partial^2 X}{\partial x^2}=0$$

$$\Rightarrow \int \frac{\partial X}{\partial x}=C_3$$

$$\Rightarrow X=C_3x+C_4$$

Putting these values in eqn. (2),

$$y=(C_1t+C_2)(C_3x+C_4)$$

Since, the wave equation is a periodic function of x & t. Therefore, it must contain trigonometrical term. So, from above it is concluded that the solution is not the solution of eqn. (1).

Case - II : When $k > 0$.

$$\frac{1}{C^2 T}\frac{\partial^2 T}{\partial t^2} = \frac{1}{X}\frac{\partial^2 X}{\partial x^2} = p^2$$

$$\Rightarrow \frac{1}{C^2 T}\frac{\partial^2 T}{\partial t^2} = p^2$$

$$\Rightarrow \frac{\partial^2 T}{\partial t^2} - p^2 C^2 T = 0$$

$$\Rightarrow (D^2 - p^2 C^2) T = 0$$

$$\Rightarrow D^2 - p^2 C^2 = 0 \quad [A.E.]$$
$$\Rightarrow D = \pm pC$$

$$\therefore T_1 = C_1' e^{pCt} + C_2' e^{-pCt}$$

Also, $\frac{1}{X}\frac{\partial^2 X}{\partial x^2} = p^2$

$$\Rightarrow \frac{\partial^2 X}{\partial x^2} = p^2 X$$

$$\Rightarrow D^2 X - p^2 X = 0$$

$$\Rightarrow (D^2 - p^2) X = 0$$

$$\Rightarrow D^2 - p^2 = 0 \quad [A.E.]$$

$\Rightarrow D = \pm p$

$\therefore X = C_3' e^{px} + C_4' e^{-px}$

Putting X & T in eq^n. (2),

$$y = (C_1' e^{pct} + C_2' e^{-pct})(C_3' e^{px} + C_4' e^{-px})$$

This is also not the solution.

Case - III : When $k < 0$.

$$\frac{1}{c^2 T} \frac{\partial^2 T}{\partial t^2} = \frac{1}{X} \frac{\partial^2 X}{\partial x^2} = -p^2$$

Taking,

$$\frac{1}{c^2 T} \frac{\partial^2 T}{\partial t^2} = -p^2$$

$$\Rightarrow \frac{\partial^2 T}{\partial t^2} + p^2 c^2 T = 0$$

$$\Rightarrow (D^2 + p^2 c^2) T = 0$$

The A.E. is,

$$D^2 + p^2 c^2 = 0$$

$$\Rightarrow D = \pm i pc$$

$T = C_1'' \cos pct + C_2'' \sin pct$

Also, $\frac{1}{X}\frac{\partial^2 X}{\partial x^2} = -p^2$

$$\Rightarrow \frac{\partial^2 X}{\partial x^2} + p^2 X = 0$$

$$\Rightarrow (D^2+p^2)X = 0$$

The A.E. is,

$$D^2+p^2 = 0$$

$$\Rightarrow D = \pm ip$$

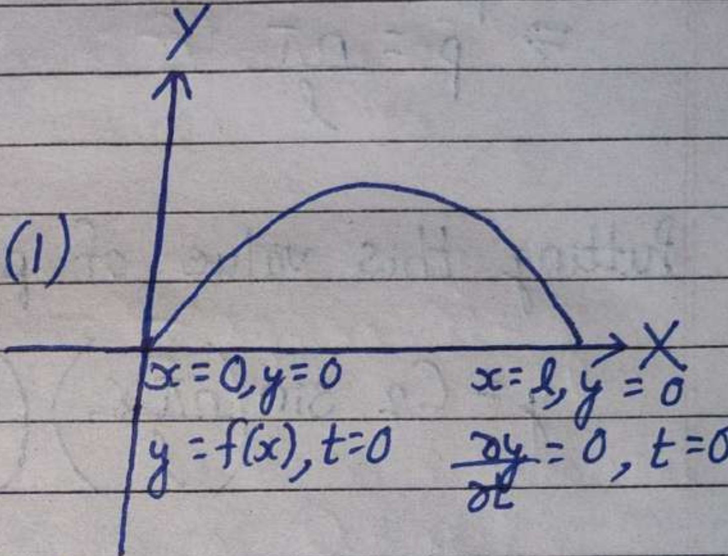
$$X = C_3'' \cos px + C_4'' \sin px$$

Putting the values of X & T in eqn. (2),

$$y = (C_1'' \cos pCt + C_2'' \sin pCt)(C_3'' \cos px + C_4'' \sin px)$$

This is the solution of the wave equation.

Q.1) Solve completely the equation $\frac{\partial^2 y}{\partial t^2} = c^2 \frac{\partial^2 y}{\partial x^2}$ representing the vibrations of a string of length l, fixed at both ends, given that $y(0,t) = 0$, $y(l,t) = 0$, $y(x,0) = f(x)$ and $\frac{\partial}{\partial t} y(x,0) = 0$, $0 < x < l$.



Sol^n:

Given P.D.E. is,

$$\frac{\partial^2 y}{\partial t^2} = c^2 \frac{\partial^2 y}{\partial x^2} \quad \text{—(1)}$$

The solution of eq^n. (1) is,

$$y = (C_1 \cos px + C_2 \sin px)(C_3 \cos Cpt + C_4 \sin Cpt) \quad \text{—(2)}$$

Now, applying the boundary condition $x=0$ & $y=0$ in eq^n. (2),

$$\cancel{0 = (C_1 + 0)(C_3 + 0)}$$

$$0 = C_1 (C_3 \cos Cpt + C_4 \sin Cpt)$$

$$\Rightarrow C_1 = 0$$

Putting this value of C_1 in eq^n. (2),

$$y = C_2 \sin px (C_3 \cos Cpt + C_4 \sin Cpt) \quad \text{—(3)}$$

Again, applying $x = l$ & $y = 0$ in eq^n. (3),

$$0 = C_2 \sin pl (C_3 \cos Cpt + C_4 \sin Cpt)$$

$$\Rightarrow \sin pl = 0$$
$$\Rightarrow \sin pl = \sin n\pi$$
$$\Rightarrow pl = n\pi$$
$$\Rightarrow p = \frac{n\pi}{l}$$

Putting this value of p in eq^n. (3),

$$y = C_2 \sin\left(\frac{n\pi x}{l}\right)\left(C_3 \cos \frac{n\pi Ct}{l} + C_4 \sin \frac{n\pi Ct}{l}\right) \quad \text{---(4)}$$

Differentiating eq^n. (4) w.r.to t, we get,

$$\frac{\partial y}{\partial t} = C_2 \sin\left(\frac{n\pi x}{l}\right)\left[-\frac{n\pi C}{l} . C_3 \sin \frac{n\pi Ct}{l} + \frac{n\pi C}{l} . C_4 \cos \frac{n\pi Ct}{l}\right]$$

Applying the condition $\frac{\partial y}{\partial t} = 0$ & $t = 0$ in above eq^n.,

$$0 = C_2 \sin\left(\frac{n\pi x}{l}\right)\left[0 + \frac{n\pi C}{l} . C_4\right]$$

$$\Rightarrow C_4 = 0$$

Putting this value of C_4 in eq^n.(4), we get,

$$y = C_2 \sin\left(\frac{n\pi x}{l}\right)\left[C_3 \cos\left(\frac{n\pi Ct}{l}\right) + 0\right]$$

$$\Rightarrow y = C_2 . C_3 . \sin\left(\frac{n\pi x}{l}\right) . \cos\left(\frac{n\pi Ct}{l}\right) \quad \text{—— (5)}$$

$$\Rightarrow y = b_n \sin\left(\frac{n\pi x}{l}\right) . \cos\left(\frac{n\pi Ct}{l}\right)$$

[where $C_2 . C_3 = b_n$]

The most general solution is,

$$y = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{l}\right) . \cos\left(\frac{n\pi Ct}{l}\right) \quad \text{—— (6)}$$

Now, applying $y = f(x)$ & $t = 0$ in eq^n.(6),

$$f(x) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{l}\right)$$

$$\Rightarrow b_n = \frac{2}{l}\int_0^l f(x) . \sin\left(\frac{n\pi x}{l}\right) dx \quad \text{—— (7)}$$

[Using Fourier Sine Series]

Hence, the required solution is eq^n.(6) and b_n is in eq^n.(7).

Q.2) If a string of length l is initially at rest in equilibrium position and each of its points is given that velocity,

$$\left(\frac{\partial y}{\partial t}\right)_{t=0} = b\sin^3\left(\frac{\pi x}{l}\right)$$

Sol^n:

The one dimension wave eq^n. is,

$$\frac{\partial^2 y}{\partial t^2} = c^2 \frac{\partial^2 y}{\partial x^2} \quad \text{——— (1)}$$

The solution of eq^n. (1) is,

$$y = (C_1 \cos px + C_2 \sin px)(C_3 \cos cpt + C_4 \sin cpt) \quad \text{——— (2)}$$

Boundary conditions,

$$y = 0 \ \& \ x = 0 \text{ ------- (i)}$$
$$y = 0 \ \& \ x = l \text{ ------- (ii)}$$
$$y = 0 \ \& \ t = 0 \text{ ------- (iii)}$$
$$\frac{\partial y}{\partial t} = b\sin^3\left(\frac{\pi x}{l}\right) \ \& \ t = 0 \text{ ------- (iv)}$$

- Applying condition $y = 0$ & $x = 0$ in eq^n. (2),

$$0 = C_1 (C_3 \cos cpt + C_4 \sin cpt)$$
$$\Rightarrow C_1 = 0$$

Putting this value of C_1 in eq^n. (2),

$$y = C_2 \sin px (C_3 \cos cpt + C_4 \sin cpt) \quad \text{——— (3)}$$

- Applying condition $y=0$ & $x=l$ in eq^n. (3),

$$0 = C_2 \sin pl (C_3 \cos cpt + C_4 \sin cpt)$$

$$\Rightarrow C_2 \sin pl = 0$$

$$\Rightarrow \sin pl = 0$$

$$\Rightarrow \sin pl = \sin n\pi$$

$$\Rightarrow pl = n\pi$$

$$\Rightarrow p = \frac{n\pi}{l}$$

Putting this value of p in eq^n. (3),

$$y = C_2 \sin\left(\frac{n\pi x}{l}\right)\left[C_3 \cos\left(\frac{n\pi ct}{l}\right) + C_4 \sin\left(\frac{n\pi ct}{l}\right)\right] \text{ —— (4)}$$

Applying the condition $y=0$ & $t=0$ in eq^n. (4),

$$0 = C_2 \sin\left(\frac{n\pi x}{l}\right) [C_3 + 0]$$

$$\Rightarrow C_3 = 0$$

Putting this value of C_3 in eq^n. (4),

$$y = C_2 \sin\left(\frac{n\pi x}{l}\right)\left[0 + C_4 \sin\left(\frac{n\pi ct}{l}\right)\right]$$

$$\Rightarrow y = C_2 . C_4 . \sin\left(\frac{n\pi x}{l}\right) . \sin\left(\frac{n\pi ct}{l}\right) \text{ —— (5)}$$

$$\Rightarrow y(x,t) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{l}\right) \cdot \sin\left(\frac{n\pi ct}{l}\right) \quad -(6)$$

Differentiating eq^n. (6) w.r.to t,

$$\frac{\partial y}{\partial t} = \sum_{n=1}^{\infty} b_n \left(\frac{n\pi c}{l}\right) \sin\left(\frac{n\pi x}{l}\right) \cdot \cos\left(\frac{n\pi ct}{l}\right)$$

Applying condition $\frac{\partial y}{\partial t} = b\sin^3\left(\frac{\pi x}{l}\right)$ & $t=0$, in above eq^n;

$$b\sin^3\left(\frac{\pi x}{l}\right) = \sum_{n=1}^{\infty} b_n \left(\frac{n\pi c}{l}\right) \cdot \sin\left(\frac{n\pi x}{l}\right) \cdot 1$$

$$\Rightarrow \frac{b}{4}\left[3\sin\left(\frac{\pi x}{l}\right) - \sin\left(\frac{3\pi x}{l}\right)\right] = \sum_{n=1}^{\infty} b_n \left(\frac{n\pi c}{l}\right) \cdot \sin\left(\frac{n\pi x}{l}\right)$$

$$\left[\begin{array}{l} \text{Using, } \sin 3\theta = 3\sin\theta - 4\sin^3\theta \\ \Rightarrow \sin^3\theta = \frac{1}{4}(3\sin\theta - \sin 3\theta) \end{array}\right]$$

$$\frac{b}{4}\left[3\sin\left(\frac{\pi x}{l}\right) - \sin\left(\frac{3\pi x}{l}\right)\right] = b_1\left(\frac{\pi c}{l}\right) \cdot \sin\left(\frac{\pi x}{l}\right) + b_2\left(\frac{2\pi c}{l}\right) \cdot \sin\left(\frac{2\pi x}{l}\right) + b_3\left(\frac{3\pi c}{l}\right) \cdot \sin\left(\frac{3\pi x}{l}\right) + \dots$$

Equating the coefficients of $\sin\left(\frac{\pi x}{l}\right)$, $\sin\left(\frac{2\pi x}{l}\right)$, $\sin\left(\frac{3\pi x}{l}\right)$, ..., we get,

$$\frac{3b}{4} = b_1\left(\frac{\pi c}{l}\right) \Rightarrow b_1 = \frac{3bl}{4\pi c}$$

$$0 = b_2\left(\frac{2\pi c}{l}\right) \Rightarrow b_2 = 0$$

$$\frac{-b}{4} = b_3\left(\frac{3\pi c}{l}\right) \Rightarrow b_3 = \frac{-bl}{12\pi c}$$

Also, $b_4 = b_5 = \ldots = 0$

Putting these values in eq^n. (6), we get,

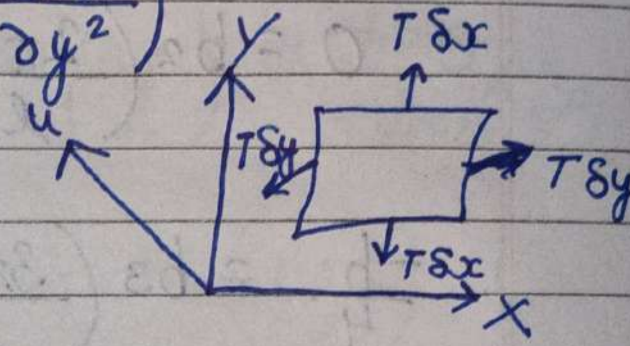
$$y = \frac{3bl}{4\pi c} \cdot \sin\left(\frac{\pi ct}{l}\right) \cdot \sin\left(\frac{\pi x}{l}\right) - \frac{bl}{12\pi c} \cdot \sin\left(\frac{3\pi ct}{l}\right) \cdot \sin\left(\frac{3\pi x}{l}\right) \quad (7)$$

Eq^n. (7) is the required solution.

Vibrating Membrane (Two Dimensional Wave Eqn.)

For a tightly stretched, uniform membrane with tension 'T' per unit length and mass 'm' per unit area, the equation for its vibrations is given by:-

$$\frac{\partial^2 u}{\partial t^2} = c^2\left(\frac{\partial^2 u}{\partial^2 x} + \frac{\partial^2 u}{\partial y^2}\right)$$



$$\left[c^2 = \frac{T}{m}\right]$$

The above eqn. is the wave eqn. in 2D.

Solution:

The two dimensional wave eqn. is,

$$\frac{\partial^2 u}{\partial t^2} = c^2\left(\frac{\partial^2 u}{\partial^2 x} + \frac{\partial^2 u}{\partial y^2}\right) \quad \text{——— (1)}$$

Let, $u = XYT$ ——— (2) be the solution of eqn. (1).

Now, $\dfrac{\partial^2 u}{\partial t^2} = XY\dfrac{d^2T}{dt^2}$

$$\frac{\partial^2 u}{\partial x^2} = YT\frac{d^2X}{dx^2}$$

$$\frac{\partial^2 u}{\partial y^2} = XT\frac{d^2Y}{dy^2}$$

Putting these values in eq^n. (1),

$$XY\frac{d^2T}{dt^2} = c^2\left(YT\frac{d^2X}{dx^2} + XT\frac{d^2Y}{dy^2}\right)$$

Dividing by c^2XYT on both sides, we get,

$$\frac{1}{c^2T}\cdot\frac{d^2T}{dt^2} = \frac{1}{X}\cdot\frac{d^2X}{dx^2} + \frac{1}{Y}\cdot\frac{d^2Y}{dy^2}$$

Choosing constants $-k^2$ & $-l^2$ such that,

$$\frac{1}{X}\frac{d^2X}{dx^2} = -k^2, \quad \frac{1}{Y}\cdot\frac{d^2Y}{dy^2} = -l^2$$

$$\& \quad \frac{1}{c^2T}\cdot\frac{d^2T}{dt^2} = -(k^2+l^2)$$

Solving each,

$$\frac{1}{X}\frac{d^2X}{dx^2} = -k^2$$

$$\Rightarrow \frac{d^2X}{dx^2} + k^2X = 0$$

$$\Rightarrow (D^2+k^2)X = 0$$

$$\Rightarrow D^2 = -k^2 \quad \text{[A. E.]}$$

$$\Rightarrow D = \pm ik \Rightarrow X = C_1\cos kx + C_2\sin kx$$

Also,

$$\frac{1}{Y}\frac{d^2Y}{dy^2} = -l^2$$

$$\Rightarrow D^2Y + l^2Y = 0$$

$$\Rightarrow (D^2+l^2)Y = 0$$

$$\Rightarrow D^2+l^2 = 0 \quad \text{[A.E.]}$$

$$\Rightarrow D = \pm il$$

$$Y = C_3 \cos ly + C_4 \sin ly$$

Again,

$$\frac{1}{c^2T}\frac{d^2T}{dt^2} = -(K^2+l^2)$$

$$\Rightarrow \frac{d^2T}{dt^2} + (K^2+l^2)c^2T = 0$$

$$\Rightarrow [D^2 + (K^2+l^2)c^2]T = 0$$

$$\Rightarrow D^2 + (K^2+l^2)c^2 = 0 \quad \text{[A.E.]}$$

$$\Rightarrow D = \pm ic\sqrt{K^2+l^2}$$

$$T = C_5 \cos(ct\sqrt{K^2+l^2}) + C_6 \sin(ct\sqrt{K^2+l^2})$$

Putting the values of X, Y & T in eq^n. (2),

$$u = (C_1 \cos Kx + C_2 \sin Kx)(C_3 \cos ly + C_4 \sin ly) \ldots$$
$$\ldots (C_5 \cos ct\sqrt{K^2+l^2} + C_6 \sin ct\sqrt{K^2+l^2}).$$

Q. Find the deflection $u(x,y,t)$ of square membrane with $a=b=c=1$, if initial velocity is zero and the initial deflection $u=f(x,y)$.

Sol^n:

The two dimensional wave eq^n. is,

$$\frac{\partial^2 u}{\partial t^2} = c^2\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\right) \text{ ——— (1)}$$

The boundary conditions are:-

$u=0,\ x=0$ ---------(i)

$u=0,\ x=1$ ---------(ii)

$u=0,\ y=0$ ---------(iii)

$u=0,\ y=1$ ---------(iv)

Initial conditions are:-

$\frac{\partial u}{\partial t} = 0,\ t=0$ --------(v)

$u=f(x,y),\ t=0$ -------(vi)

~~Now,~~ the solution of eq^n. (1) is,

$$u=(C_1\cos kx + C_2\sin kx)(C_3\cos ly + C_4\sin ly)(C_5\cos ct\sqrt{k^2+l^2} + C_6\sin ct\sqrt{k^2+l^2}) \text{ ——— (2)}$$

Applying (i) boundary condition in eq^n. (2), we get,

$\to (u=0\ \&\ x=0)$

$C_1 = 0$

Putting this value of C_1 in eq^n. (2),

$$u = C_2 \sin kx \,(C_3 \cos ly + C_4 \sin ly)\,(C_5 \cos ct\sqrt{k^2+l^2} + C_6 \sin ct\sqrt{k^2+l^2}) \quad \text{---(3)}$$

Applying (ii) condition i.e. $u=0$ & $x=1$ in eq^n. (3),

$$\sin k = 0$$
$$\Rightarrow \sin k = \sin m\pi$$
$$\Rightarrow k = m\pi$$

Putting this value of k in eq^n. (3),

$$u = C_2 \sin m\pi x \,(C_3 \cos ly + C_4 \sin ly)\,(C_5 \cos ct\sqrt{m^2\pi^2+l^2} + C_6 \sin ct\sqrt{m^2\pi^2+l^2}) \quad \text{---(4)}$$

Applying condition (iii) i.e. $u=0$ & $y=0$ in eq^n. (4),

$$C_3 = 0$$

Putting this value of C_3 in eq^n. (4),

$$u = C_2 C_4 \sin m\pi x \,.\, \sin ly \,(C_5 \cos ct\sqrt{m^2\pi^2+l^2} + C_6 \sin ct\sqrt{m^2\pi^2+l^2}) \quad \text{---(5)}$$

Applying condition (iv) i.e. $u=0$ & $y=1$ in eq^n. (5),

$$\sin l = 0 = \sin n\pi$$
$$\Rightarrow l = n\pi$$

Putting this value of l in eq^n. (5),

$$u = C_2 C_4 \sin m\pi x . \sin n\pi y \left(C_5 \cos ct\sqrt{m^2\pi^2+n^2\pi^2} + C_6 \sin ct\sqrt{m^2\pi^2+n^2\pi^2}\right)$$

$$\Rightarrow u = C_2 C_4 . \sin m\pi x . \sin n\pi y \, (C_5 \cos pt + C_6 \sin pt) \quad \text{—— (6)}$$

[where $p = \pi c\sqrt{m^2+n^2}$]

Applying condition (v) i.e. $\partial u/\partial t = 0$ & $t = 0$ in eq^n. (6),

$$C_6 = 0$$

Putting this value of C_6 in eq^n. (6),

$$u = C_2 C_4 C_5 . \sin(m\pi x) . \sin(n\pi y) . \cos(pt)$$

$$\Rightarrow u = B_{mn} . \sin(m\pi x) . \sin(n\pi y) . \cos(pt).$$

or the most general solution is,

$$u = \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} B_{mn} . \sin(m\pi x) . \sin(n\pi y) . \cos(pt) \quad \text{—— (7)}$$

Applying the condition (vi) i.e. $u = f(x,y)$ & $t = 0$ in eq^n. (7),

$$f(x,y) = \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} B_{mn} . \sin(m\pi x) . \sin(n\pi y)$$

By the double fourier sine series,

$$B_{mn} = \frac{2}{1} \cdot \frac{2}{1} \int_0^1 \int_0^1 f(x,y) \cdot \sin(m\pi x) \cdot \sin(n\pi y)\, dx\, dy$$

$$\Rightarrow B_{mn} = 4 \int_0^1 \int_0^1 f(x,y) \cdot \sin(m\pi x) \cdot \sin(n\pi y)\, dx\, dy \quad \text{——— (8)}$$

Hence, the required solution is eqⁿ. (7) and the eqⁿ. (8) gives B_{mn}.